

HAT-P-12b

R-0783

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-12_230417 · created 2026-07-28T06:31:59+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2460051.6851 +/- 0.003 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.157 +/- 0.013
Transit depth (Rp/Rs)^2	2.45 +/- 0.39 [%]
Orbital Inclination (inc)	88.05 +/- 0.93
Airmass coefficient 1 (a1)	0.953 +/- 0.014
Airmass coefficient 2 (a2)	0.043 +/- 0.013
Scatter in the residuals of the lightcurve fit is	1.51 %
Best Comparison Star	#1 - [278, 115]
Optimal Aperture	3.85
Optimal Annulus	9.89
Transit Duration (day)	0.092 +/- 0.0075

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.157 ± 0.013 vs 0.1406 (deviation 1.24σ)
Duration fit vs published	0.092 d vs 0.0975 d (deviation 0.72σ)
Mid-time O–C	-2.2 min
Depth SNR	11.9
Residual scatter	1.51 %

Light curve

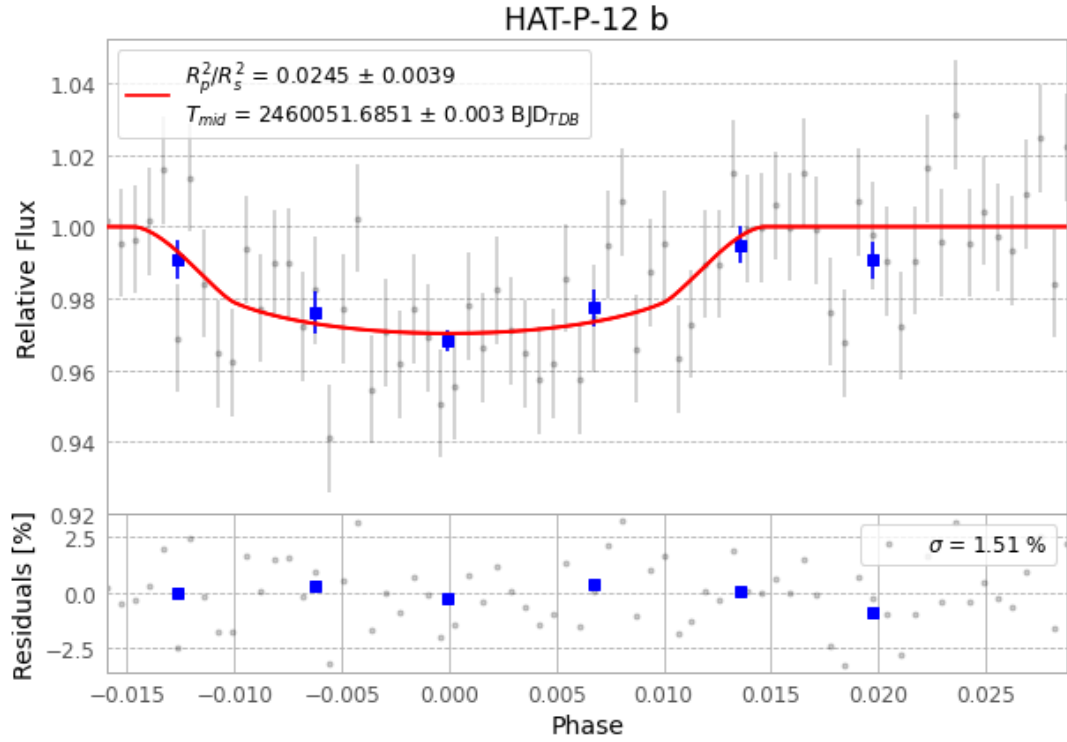


Figure 1 — FinalLightCurve_HAT-P-12 b_2023-04-16.png (SHA-256 003ab4a6175e6482...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [351, 35], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 1c945f2684ebdc22...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

70 FITS frames (46 MB), HATP-12230417030615.FITS → HATP-12230417063315.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-12-230417.log (2122639a9a41...), FinalLightCurve_HAT-P-12 b_2023-04-16.png (003ab4a6175e...), FinalLightCurve_HAT-P-12 b_2023-04-16.csv (a962977a9752...)

Compute resources

Wall time 105.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)